QFT 2 Lecture Note

Yu Liu

April 16, 2026

Abstract

This is a lecture note of the QFT 2 course in EPFL

Contents

1 Lecture 1: Massless Representations of the Poincaré Group

We have classified the Unitary Irreducible Representations (UIRs) of the Poincaré group into two classes: massive representations and massless representations. In this lecture, we will focus on the massless representations.

1.1 Recap of UIR Construction

To construct a UIR of Poicare group we need to construct it for both translation and Lorentz transformation, represented as:

$$U(a) = e^{i a_\mu P^\mu}, \quad U(\Lambda) = e^{-\frac{i}{2}\omega_{\mu\nu} J^{\mu\nu}} \tag{1}$$

To do this we assume a UIR space with basis $|p, \sigma\rangle$, which diagonalizes the operator P^μ. Then we can construct the UIR of translation by:

- For Translation: $U(a)\ |p, \sigma\rangle = e^{i a_\mu p^\mu}\ |p, \sigma\rangle$.
- For Lorentrz transformation we need to use **Wigner's Little Group**
  - Step 1: define a "Standard Boost" and assume it satisfy: $U(\Lambda_p)\ |k, \sigma\rangle = |p, \sigma\rangle$
  - Step 2: for a general lorentz transformation is given by a combination of wigner little group and the standard boost. We only need to consider a representation of the wigner little group, which is given by: $W(\Lambda, p) = \Lambda_{\Lambda p}^{-1} \Lambda \Lambda_p$.

We know that the generator of Wigner's little group is exactly given by the **Pauli-Lubanski** vector. Thus, long as we use a irreducable representation of the Wigner's little group, we can get a UIR of the Lorentz transformation.

$$\begin{aligned} U(\Lambda)\ |p, \sigma\rangle &= H(\Lambda_p) U(W(\Lambda, p))\ |k, \sigma\rangle \\ &= \sum_{\sigma'} D_{\sigma' \sigma}(W(\Lambda, p))\ |\Lambda p, \sigma'\rangle \end{aligned} \tag{2}$$

1.1.1 Helicity Basis

Different Basis corresponds to different "standard boost". For helicity basis, we choos the standard boost to be:

$$\Lambda_p^h = R(\hat{p}) e^{i K_3 \eta_p}, \quad \eta_p = \operatorname{arctanh}\left(\frac{|\vec{p}|}{p^0}\right) \tag{3}$$

which is:

- first boost to the speed of p along the z-axis, and then
- rotate to the direction of p.

Note that here K_3 is in the defining representation of Lorentz group.

1.2 Massless Representations

1.2.1 Wigner Little Group

For a massless representation, we take the standard momentum to be $k^\mu = (k, 0, 0, k)$. We can directly get the Wigner's Little group by calculating the Pauli-Lubanski vector, which is given by:

$$W^0 = -k J^3, \quad W^1 = k(K^2 - J^i), \quad W^2 = k(-J^2 - K^1), \quad W^3 = -k J^3 \tag{4}$$

We can see that the Wigner's Little group only have 3 generators satisfying the commutation relation of:

$$[J^3, W^1] = iW^2, \quad [J^3, W^2] = -iW^1, \quad [W^1, W^2] = 0 \tag{5}$$

This is the algebra of ISO(2) group.

1.2.2 Representation of Wigner's Little Group

We now try to find a representation for this algebra. We can reconbine the generators as:

$$W^{\pm} = W^1 \pm iW^2 \tag{6}$$

Then the commutation relation becomes:

$$[J^3, W^+] = W^+, \quad [J^3, W^-] = -W^-, \quad [W^+, W^-] = 0 \tag{7}$$

We can see that $W^{\pm}$ behaves like the raising and lowering operator, but the noncommutative nature tells us that state generated by $W^{\pm}$ are not normalizable. We can further see this from below

To construct a representation for this algebra we can have $|k, \lambda\rangle$ as the basis of the representation space which is an eigenstate of J^3 with eigenvalue λ. Then we can see that:

$$J^3 W^+ |k, \lambda\rangle = (\lambda + 1) W^+ |k, \lambda\rangle, \quad J^3 W^- |k, \lambda\rangle = (\lambda - 1) W^- |k, \lambda\rangle \tag{8}$$

And the Casimir of the algebra is given by:

$$W^2 = -W^+ W^- \tag{9}$$

For an irreducible representation, the Casimir should be proportional to the identity operator, which means that $W^+ W^- = -c\mathbb{I}$. Moreover, we note that by definition and the restriction of representation to be unitary, we have: $W^- = (W^+)^\dagger$. Thus we can see that c should be non-negative, and we can have two cases:

- $c > 0$: then we have $(W^-)^\dagger W^- \propto 1$, which means that the lowering operator of the algebra is a unitary operator. Thus the representation space is infinite dimensional, which is not physical.
- $c = 0$: then we have $W^+ = W^- = 0$ and the representation space is one dimensional, which is physical. Thus this is the case we are interested in。

In this case, the representation of the Wigner's Little group is one dimensional and labelled by λ. The representation is given by:

$$W^\mu |k, \lambda\rangle = -\lambda k^\mu |k, \lambda\rangle \tag{10}$$

We can generalize this result to a general p by using the commutation relation of $W^m u$ and Lorentz Generators. The results is:

Theorem 1.1 **Massless Representation of Wigner's Little Group**

For a massless representation of the Poincaré group, the representation of Wigner's Little group is one dimensional and labelled by λ. The representation of generators are given by:

$$W^\mu |p, \lambda\rangle = -\lambda p^\mu |p, \lambda\rangle, \quad W^\mu = -\lambda P^\mu \tag{11}$$

we can prove this by:

$$H(p) W^\mu H(p)^\dagger H(p) |k, \lambda\rangle = H(p) W^\mu |k, \lambda\rangle = -\lambda H(p) k^\mu |p, \lambda\rangle \tag{12}$$

and we have the commutation relation:

$$H(p)W^{\mu}H(p)^{\dagger} = W^{\nu}\Lambda^{\mu}_{p\nu} \tag{13}$$

1.2.3 Interpretation of λ

We can give λ a physical interpretation. We choose the standard boost as:

$$\Lambda^{h}_{p} = R(\hat{p})e^{iK_3\eta_p}, \quad \eta_p = \operatorname{arctanh}\left(\frac{|\vec{p}|}{p^0}\right) \tag{14}$$

We know from Equation 12 that the state $|p,\lambda\rangle$ is an eigenstate of $H(p)J^3H(p)^{\dagger}$ eigenvector with eigenvalue λ. In the helicity basis, we plug in the form of $H(p)$ we have:

$$H(p)J^3H(p)^{\dagger} = R(\hat{p})e^{iK_3\eta_p}J^3e^{-iK_3\eta_p}R(\hat{p})^{\dagger} = R(\hat{p})J^3R(\hat{p})^{\dagger} = J\cdot\hat{p} \tag{15}$$

We thus see that the basis vector $|p,\lambda\rangle$ in the helicity basis is an eigenvector of the operator:

$$J\cdot\hat{p}\ |p,\lambda\rangle = \lambda\ |p,\lambda\rangle \tag{16}$$

Thus we interpret λ as the helicity of the state, which is the projection of the spin along the direction of motion.

In fact, the form of $H(p)W^{\mu}H(p)^{\dagger}$ is independent of the choice of explicit form of $H(p)$ for **massless representation**. Yet, helicity basis is good for directly showing this result.

1.2.4 Helicity Basis UIR

We explicitly write down the "standard boost" for the helicity basis:

$$\Lambda^{h}_{p} = R(\hat{p})e^{iK_3\eta_p} \tag{17}$$

where

$$e^{iK_3\eta_p} = \begin{pmatrix} \cosh(\eta_p) & 0 & 0 & \sinh(\eta_p) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh(\eta_p) & 0 & 0 & \cosh(\eta_p) \end{pmatrix}, \quad \eta_p = \operatorname{arctanh}\left(\frac{|\vec{p}|}{p^0}\right) \tag{18}$$

and $R(\hat{p})$ is standard rotation. If we now assume that $H(p) = U(\Lambda^{h}_{p})$ that gives us:

$$H(p)\ |k,\lambda\rangle = |p,\lambda\rangle \tag{19}$$

as the LT that doesn't change the λ index. Now we construct the full UIR in Helicity basis, as Equation 2, we have in Helicity basis:

$$U(\Lambda)|p,\lambda\rangle = H(\Lambda p)U(W(\Lambda,p))\ |k,\lambda\rangle \tag{20}$$

and we know that $U(W(\Lambda,p))$ is generated by only one non-zero generator J^3 for the $c=0$ case, thus, we assume the lie parameter is given by $\theta(\Lambda,p)$ which can be explicitly calculated from the definition:

$$W(\Lambda,p) = \Lambda^{h-1}_{\Lambda p}\Lambda\Lambda^{h}_{p} = e^{i\theta(\Lambda,p)J^3} \tag{21}$$

(in the defining representation of lorentz group) Then we have:

$$U(\Lambda)\ |p,\lambda\rangle = e^{i\theta(\Lambda,p)\lambda}\ |\Lambda p,\lambda\rangle \tag{22}$$

This is the representation of the Lorentz transformation for massless representation in the helicity basis.

Theorem 1.2 **Massless Representation of Poincare group in Helicity Basis**

For a massless representation of the Poincaré group, the representation of the Lorentz transformation in the helicity basis is given by:

$$U(\Lambda)|p,\lambda\rangle = e^{i\theta(\Lambda,p)\lambda}\ |\Lambda p,\lambda\rangle \tag{23}$$

the representation of the translation is given by:

$$U(a)|p,\lambda\rangle = e^{ia_\mu p^\mu}\ |p,\lambda\rangle \tag{24}$$

1.2.5 Values of λ

Just as the label of massive representation the spin j can only take integer or half-integer values, the helicity λ of massless representation also have restrictions. The restriction is from the global topology of the lorentz group:

- Any 4π rotation must give back the identity element.

Thus we have:

$$e^{i4\pi\lambda} = 1, \quad \lambda \in \frac{Z}{2} \tag{25}$$

Moreover, we want the state not only to be representation of **Poincare group** but also form a representation of **CPT** transformation. This shows that a single λ representation is not enough, we need to have both λ and $-\lambda$ representation to form a representation of CPT transformation.

Thus we have the final result that the helicity λ can only take integer or half-integer values, and for each λ we also need to have a $-\lambda$ representation.

For example, for photon, we have $\lambda = 1$ and $\lambda = -1$ representation.

2 Lecture 2: Classical U(1) Gauge Theory

2.1 Classical Vector Field with U(1) Gauge Symmetry

2.1.1 Lagrangian of U(1) Gauge Theory

We want to construct a theory of vector fields, but more than that we require the theory to have a U(1) gauge symmetry, which means that:

$$A_{\mu(x)} \to A_{\mu(x)} - \partial_\mu \alpha(x) \tag{26}$$

is a local symmetry of the theory for any function $\alpha(x)$. We can show that the only Lagrangian that is invariant under this gauge symmetry is:

Theorem 2.1 **Lagrangian of U(1) Gauge Theory**

The Lagrangian of a U(1) gauge theory is given by:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \tag{27}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the field strength tensor.

2.1.2 DoF Counting

We now count the degree of freedom of the theory. Naively, we can see A_μ is a real 4-vector, which has 4 DoF. However, we have a gauge symmetry, which means that we can use the gauge symmetry to fix one of the DoF. Thus, we have 3 DoF left.

This can also be seen if we view the field strength tensor $F_{\mu\nu}$ as the fundamental DoF of the theory. We can see that $F_{\mu\nu}$ is an antisymmetric 2-tensor, which has 6 independent components. However, we have the Bianchi identity:

$$\varepsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0 \tag{28}$$

which gives finally 3 independent DoF for the field strength tensor, which is consistent with the DoF counting from the gauge field A_μ.

2.1.3 Interaction with Other Fields

We now consider all possible interactions between the gauge field A_μ and other fields.

- Choice 1: $F_{\mu\nu}$ coupling, frist we can consider following interaction:

$$F_{\mu\nu} F^{\mu\nu} \varphi, \quad F_{\mu\nu} \bar{\psi} \sigma^{\mu\nu} \psi \tag{29}$$

However, these interaction have an explicit $F_{\mu\nu}$, one can prove that these are non-renormalizable.

- Choice 2: A_μ coupling. Yet we may face the problem that A_μ is not gauge invariant, thus we need to define a gauge transformation for the other fields to make the interaction gauge invariant. There are many ways to do this:
  - Dirac Field: we can define the gauge transformation for a Dirac field ψ as:

$$\psi(x) \to e^{iq\alpha(x)} \psi(x) \tag{30}$$

  - Complex Scalar Field: we can define the gauge transformation for a complex scalar field φ as:

$$\varphi(x) \to e^{iq\alpha(x)} \varphi(x) \tag{31}$$

- Real Scalar Field: we can define the gauge transformation for a real scalar field φ as:

$$\varphi(x) \to \varphi(x) + M\alpha(x) \tag{32}$$

With these gauge transformation, we can construct the following gauge invariant interaction with a tool called covariant derivative:

Definition 2.2 **Covariant Derivative**

The covariant derivative is defined as a derivative transformed under the gauge transformation as:

$$D_\mu \to U(x) D_\mu U(x)^{-1} \tag{33}$$

for U(1) gauge theor, we have $U(x) = e^{iq\alpha(x)}$, thus we have:

$$D_\mu = \partial_\mu + iqA_\mu \tag{34}$$

Thus we can couple by using the covariant derivative, for example:

$$\bar{\psi}\gamma^\mu D_\mu \psi, \quad (D_\mu \varphi)^* D^\mu \varphi \tag{35}$$

2.1.4 Coupling to U(1) Conserved Current

We then can see an interesting fact if we single out the interaction term, we can see that for the Dirac field, we have:

$$\mathcal{L}_{\text{int}} = -q\bar{\psi}\gamma^\mu A_\mu \psi \tag{36}$$

and for the complex scalar field, we have:

$$\mathcal{L}_{\text{int}} = -iq(\varphi^* \partial^\mu \varphi - \varphi \partial^\mu \varphi^*) A_\mu + q^2 A_\mu A^\mu \varphi^* \varphi \tag{37}$$

We may see a fact that **the U(1) gauge field couple to the global U(1) current**. This is not a coincidence. In fact, in we want to lift a global U(1) symmetry to a local U(1) symmetry, we know that the viration of the matter field Lagrangian under the global U(1) transformation is given by:

$$\delta \mathcal{L}_m = J_N^\mu \partial_\mu \alpha(x) \tag{38}$$

Then if we want the full Lagrangian with the gauge field to be invariant under the local U(1) transformation, we need to add a term to cancel the variation. We assume that the full Lagrangian is given by:

$$\mathcal{L} = \mathcal{L}_m + \mathcal{L}_{\text{gauge}} + A^\mu J_\mu \tag{39}$$

Then:

$$\delta \mathcal{L} = J_N^\mu \partial_\mu \alpha(x) + A^\mu \delta J_\mu + \delta A^\mu J_\mu = 0 \tag{40}$$

we know that $\delta A^\mu = -\partial^\mu \alpha(x)$, thus if $\delta J_\mu = 0$ under the gauge transformation (Dirac Field Case), we can see that the symmetry condition givs us:

$$J_N^\mu = J^\mu \tag{41}$$

Thus we can interpret the global U(1) symmetry conserved current as **electric current**.

Remark

Note that $\delta I_\mu = 0$ is only true for dirac field. For scalar field, the current is not invariant under the gauge transformation, thus we have $\delta J_\mu \neq 0$, which cause to the extra term.

2.2 U(1) Gauge Field EoM (aka Maxwell's Equation)

3 Lecture 3: U(1) Gauge Theory Quantization

4 Lecture 4: Massive Vector Field

5 Lecture 5: Cauality in QFT

6 Lecture 6: Scattering Theory

Here we basically follows Weinberg's approach to scattering theory. In fact the correct understanding of this should be based on the LSZ reduction formula, but we will not go into that in this lecture.

6.1 In and Out States

To set up the scattering theory, we need to define the in and out states.

6.1.1 Free Partical States

In free QFT

7 Lecture 7: Scattering Theory II: Symmetry

8 Lecture 8: Scattering Theory III: Cross Section and Decay Rate

Now we focus on how to compute some physical quantities from the S matrix of a Scattering Theory. This discussion mainly follows the approach in Peskin and Schroeder's book [1].

8.1 Differential Cross Sections

8.2 DCS from S Matrix

Now we discuss how to get

8.2.1 Probablity Density

An naive obeservable is the propablility of event happening. To describe probability in QM we use a **projection measuring**:

$$P(\psi \to \Omega) = \langle \psi | \mathbb{P}_\Omega | \psi \rangle \tag{42}$$

In QFT we mainly care about the probablity of a final state being generated under given initial condition of the scattering. A final state (for example for spinless particles) can be described as a free many particle state with certain momenta and spins:

$$|p_1, p_2, ..., p_n\rangle \tag{43}$$

Then for exampl

In reality, our scattering set up may not be able to fix

8.2.2 Differential Event Number

8.2.3 Differential Cross Section

Bibliography

[1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*. Reading, Mass: Addison-Wesley Pub. Co, 1995.